FRACTAL TREES

Task 1 CAT Investigation

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WHAT IS A FRACTAL TREE?

What are the similarities between a fractal tree and a real tree?

Similarities/Realistic:

- It has a trunk on the bottom that splits into branches like real trees
- The branches keep splitting into smaller branches the higher you go up
- It spreads out wider at the top much like a real tree

Diferences/Unrealistic:

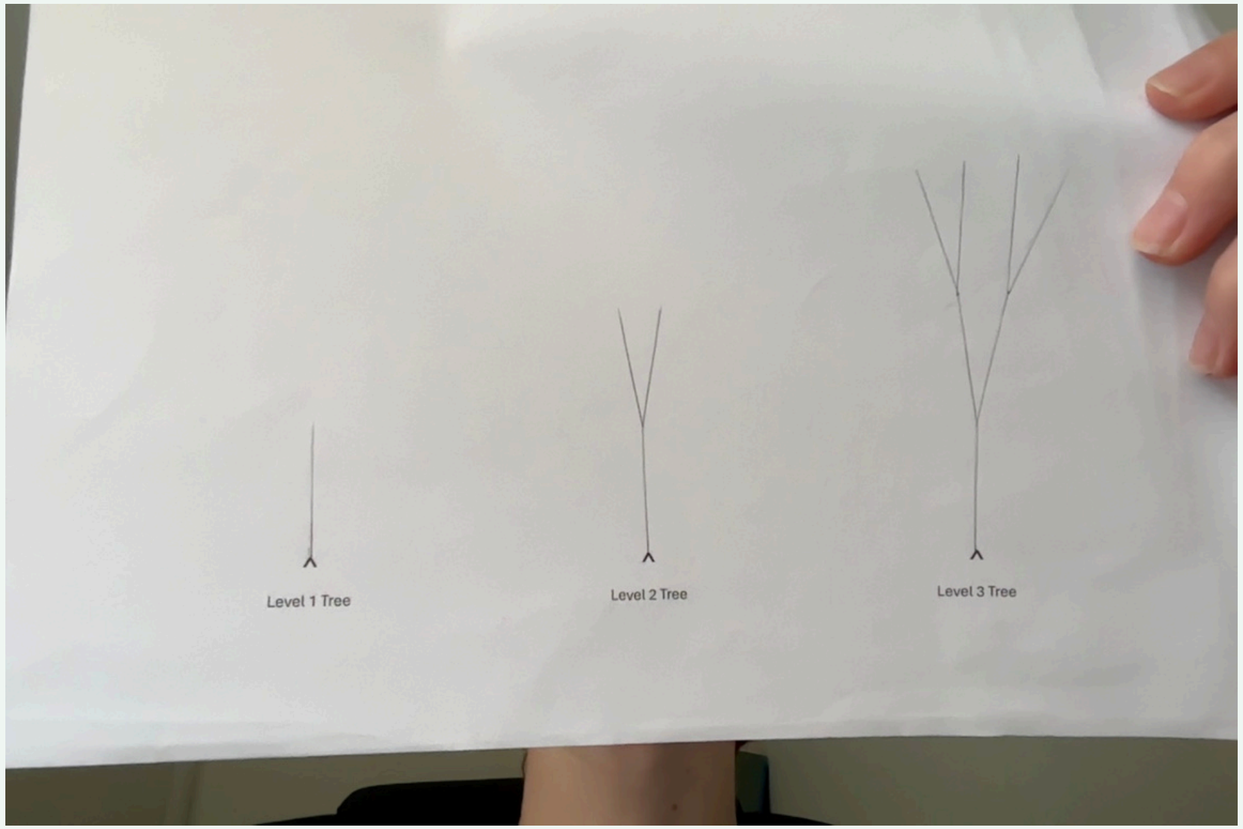
- Every branch is exactly the same length when actual trees vary a lot
- Its perfectly symmetrical, when real trees are uneven and ireegular
- There are no leaves
- The angles between the branches are all exactly the same

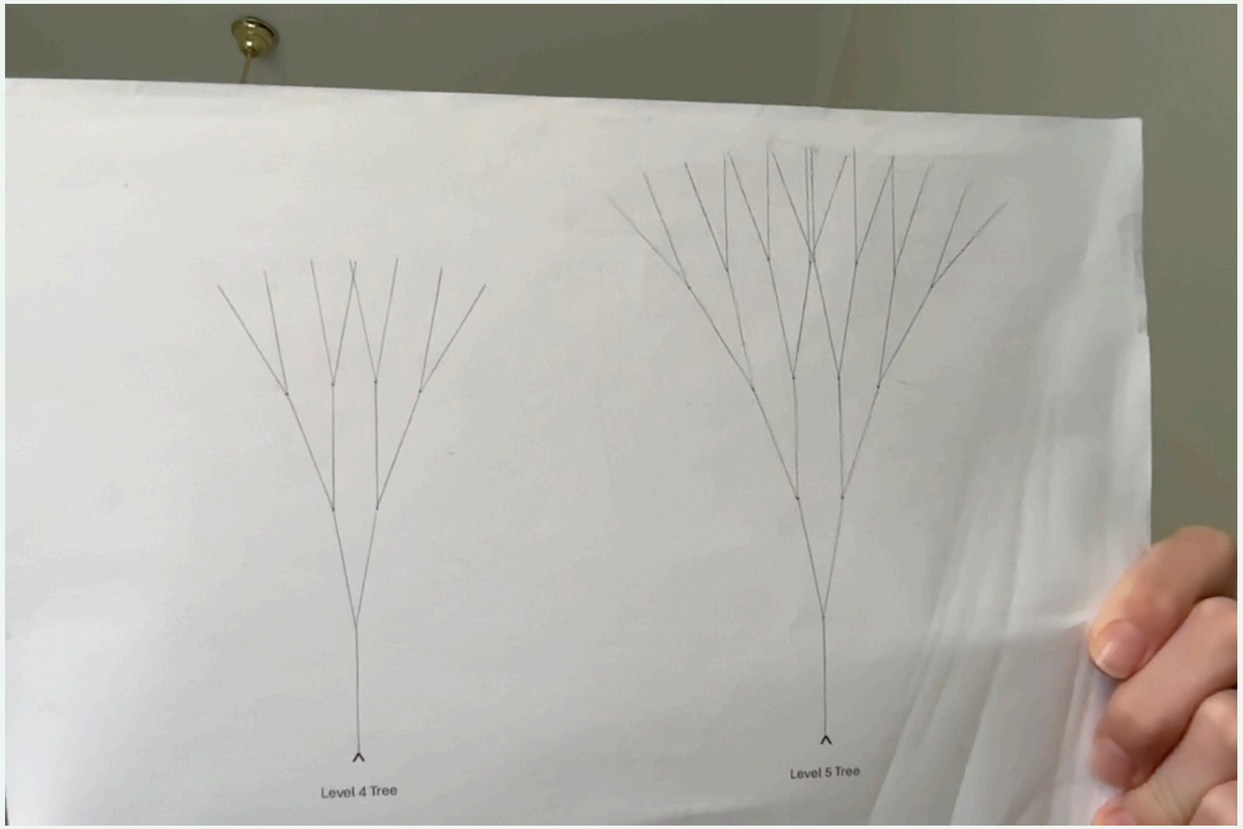
HOW SELF SIMILARITY IS USED TO DRAW TREES OF DIFFERENT LEVELS

Fractal structure explanation:

A fractal tree is “self similiar” meaning that if you look at one branch or “path” it looks like a mini version of the whole tree. This is why the instructions work the way they do because each level just repeats the exact same steps but starting from a different branch. So to draw a level five tree you just draw a level 4 tree but one more time. So you just add 2 branches to each branch which is self repeating.

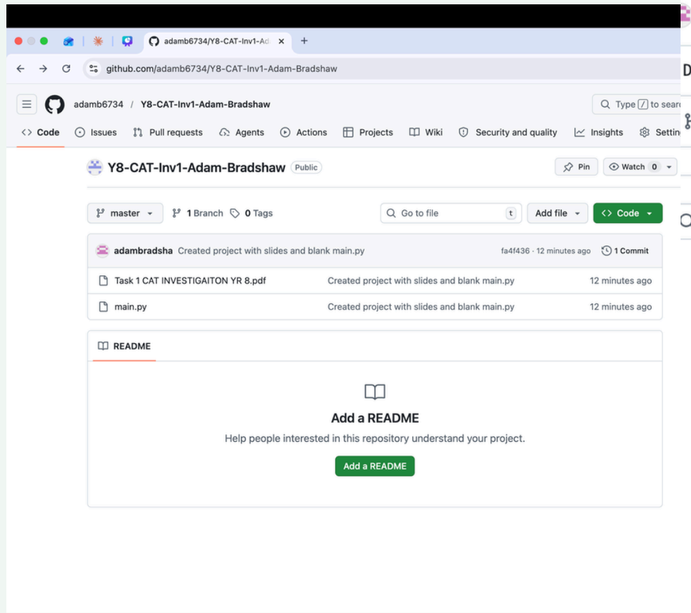
THE TREES I DREW

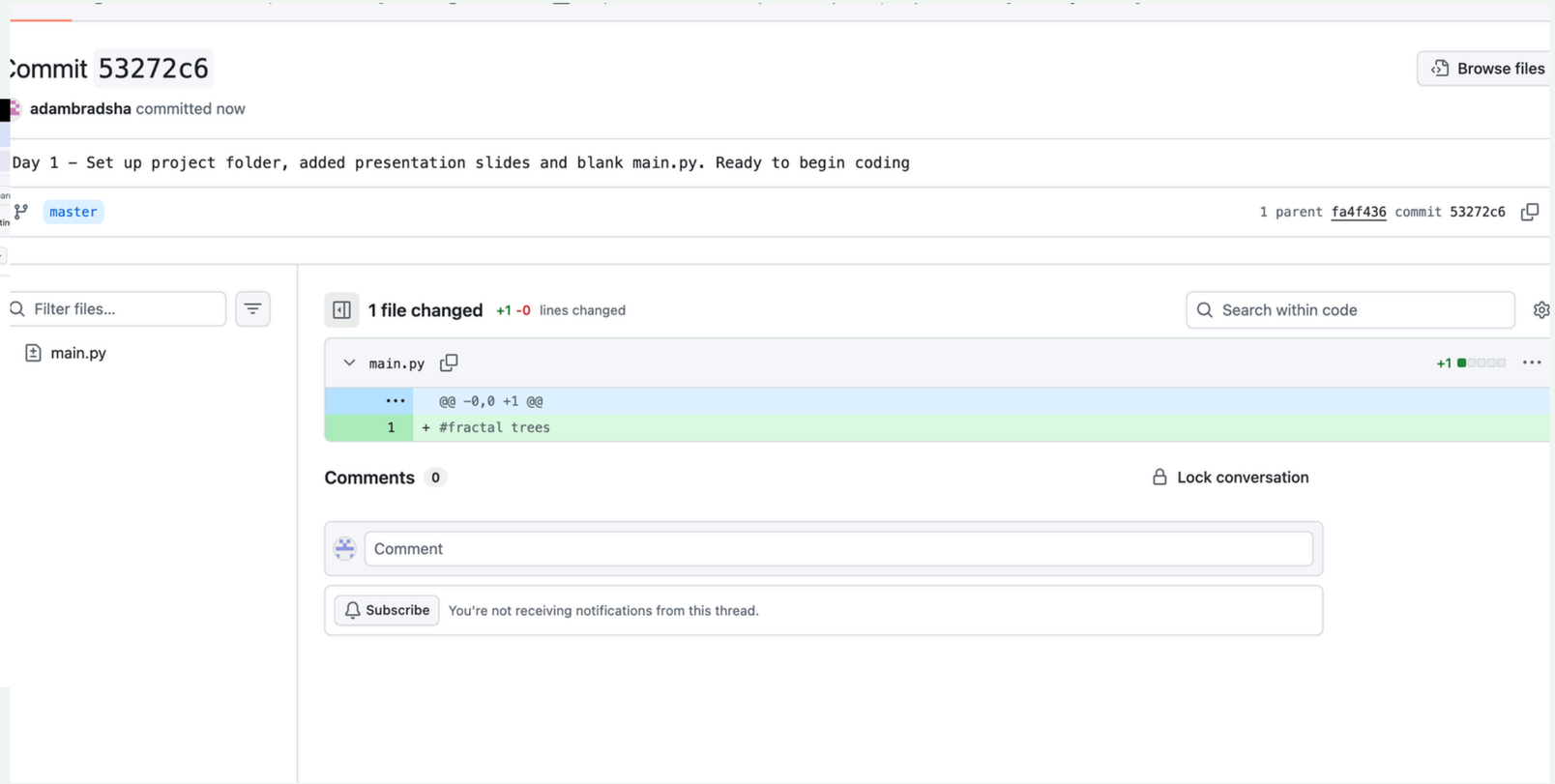




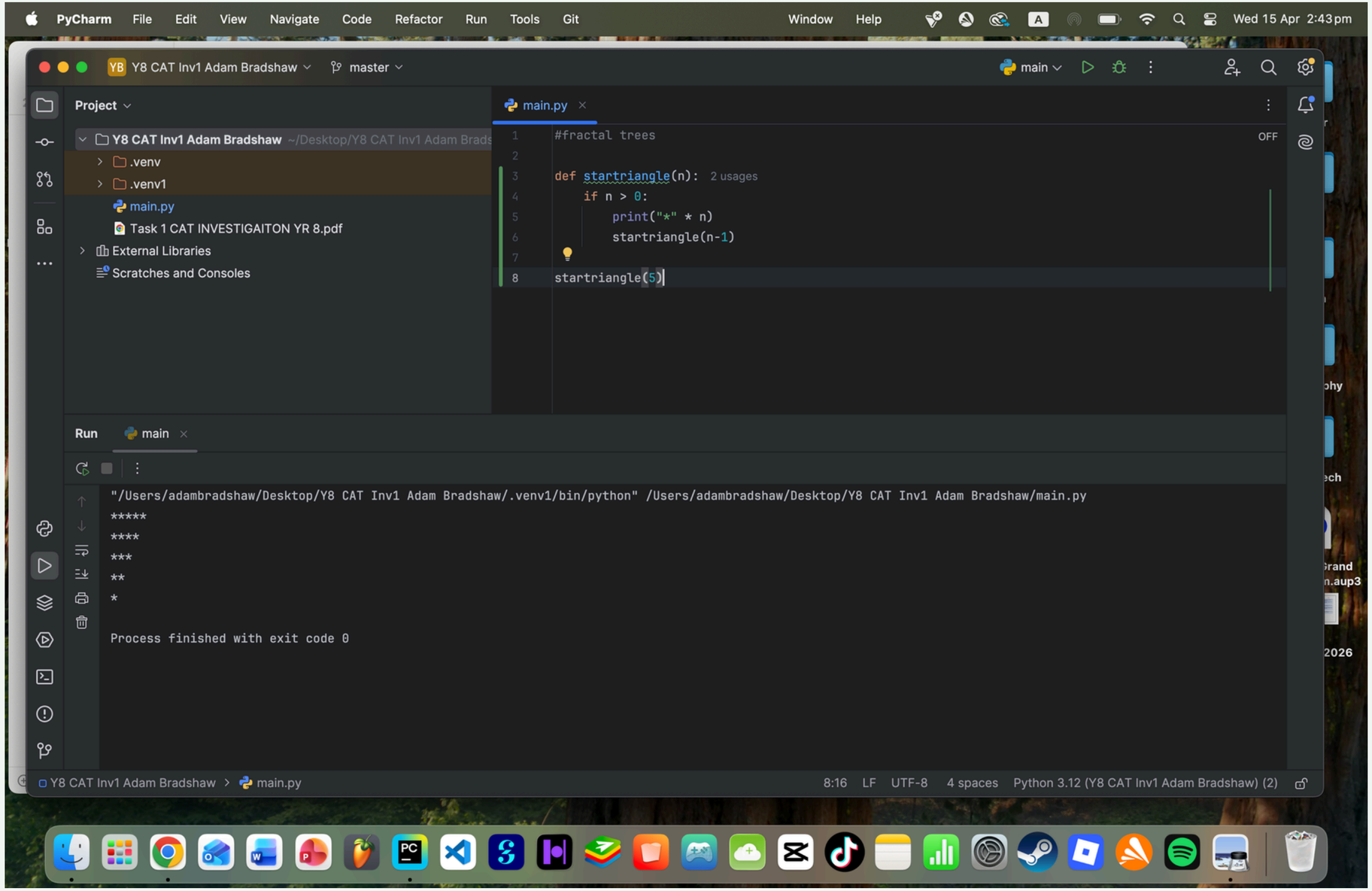
Levels 1-3 (The angle that i chose to put them at was 10°) Levels 4-5

Task 3 SETTING UP THE PROJECT





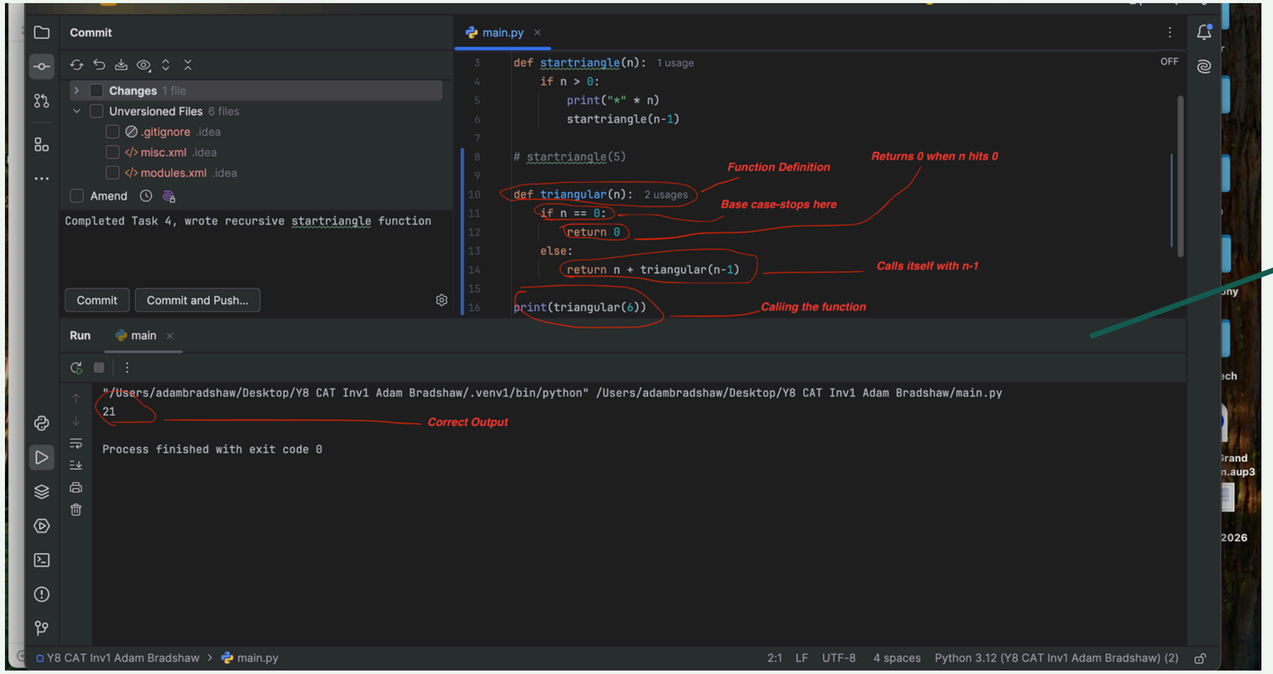
Task 4 Startriangle Function:



This recursive function prints one less star each time it calls itself until it reaches 0

TASK 5

Triangular Number Function

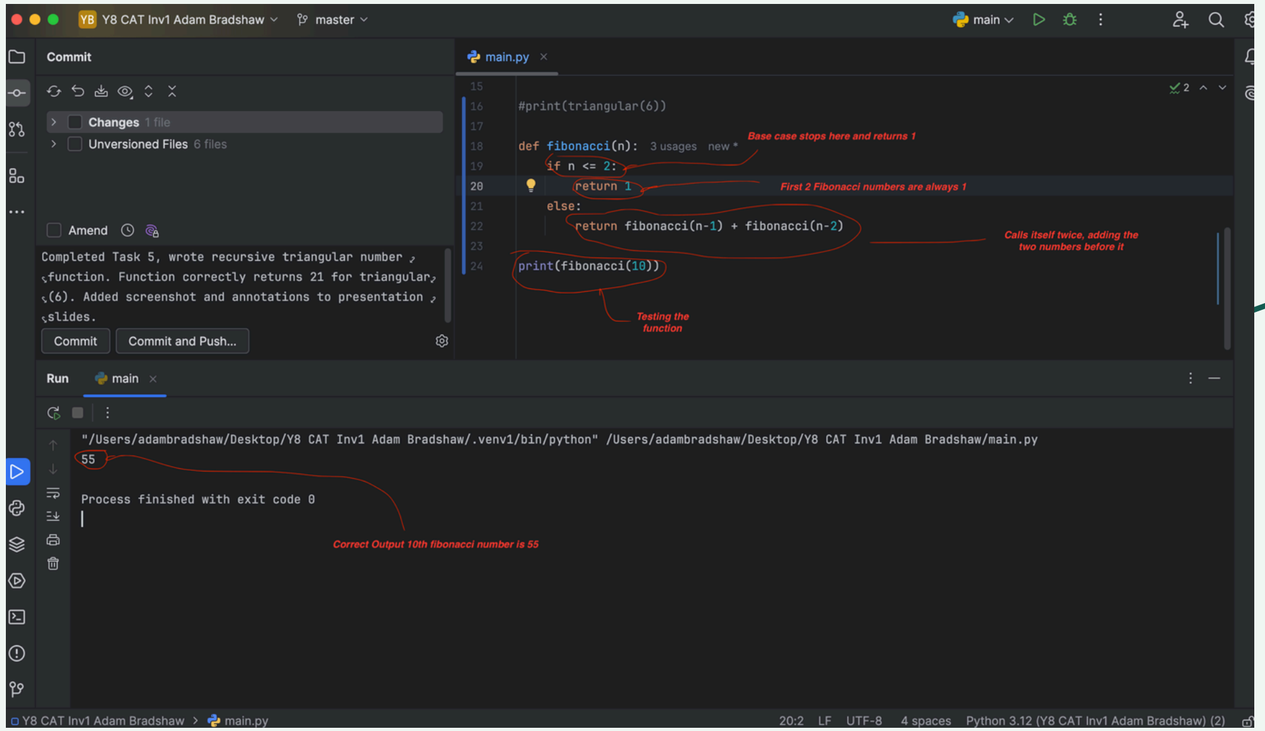


Annotations inside the screenshot

This recursive function adds n to the result of calling itself with n-1 until it reaches 0

TASK 6

Fibonacci Function



Annotations inside the screenshot

This recursive function returns the nth Fibonacci number by adding the two previous numbers together until it hits the base case